F1 Pit Stop DL Project Final Report

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Abstract

Formula 1 (F1) racing relies heavily on optimal pit-stop timing to minimize overall race duration, yet training robust predictive deep learning models for this task is hindered by the limited availability of highly sequential, real-world telemetry data. To address this fundamental data scarcity, we frame the pit-stop decision as a multivariate time-series binary classification task. Our core approach utilizes a multi-layer Long Short-Term Memory (LSTM) network for lap-by-lap decision making, which is augmented by a novel synthetic data generation pipeline utilizing Time-LLM. By applying Parameter-Efficient Fine-Tuning (specifically, LoRA) to a foundational LLaMA model, we reprogram the language model to understand and forecast continuous numerical telemetry patches rather than text. Our baseline LSTM model demonstrated strong initial performance, achieving 92.0% accuracy and a 0.92 F1-score on a real-world test split taken from the OpenF1 dataste. Additionally, by augmenting our training set with physics-validated synthetic telemetry generated by our Time-LLM pipeline, our final model achieved an improved accuracy of 93.43% and an F1-score of 0.94, a slight improvement of about 2%. This demonstrates that reprogramming text-native LLMs for continuous sensor data can effectively robustify sequential decision models in complex, data-constrained motorsport environments.

1 Introduction

Post 2020, Formula 1 (F1) Racing has been rising in popularity. F1, often referred to as ”the pinnacle of motorsport” Honda Motor Co., Ltd. (2026), usually involves 10 teams and 20 cars, with prospective expansions to 11 teams and 22 cars. All of these cars have one goal: to minimize their lap time around different tracks across the globe.

In F1, a pit stop is not merely a routine maintenance task; it is a mandatory regulatory requirement, as drivers must use at least two different tire compounds during a dry race. Beyond compliance, pit stops serve as a critical strategic battleground that often differentiates a win from a loss. Pit stops are essential to racing, allowing the pit crew to swap out tires and make aerodynamic adjustments in order to push their team further to victory, with the average pit stop taking 2-2.5 seconds Stavropoulos (2018). Teams use pit stops as a strategy in and of itself; for example, timing a pit stop slightly earlier or later than a rival (an ”undercut” or ”overcut”) may allow a team to gain a massive time advantage and overtake their position. Therefore, many dynamic factors go into whether to pit.

Our project aims to understand the optimal pit stop for all teams based on empirical race data. Our objective is to build a binary classifier that predicts whether a driver will pit on the next lap based on real-time sequential data recorded from the races. To guide this, we frame our work around a crisp research question: *Can Large Language Models (LLMs) be reprogrammed to synthesize realistic time-series telemetry to augment and improve the generalizability of an LSTM-based pit-stop classifier?*

We define our success criteria primarily through the F1-score and Recall metrics evaluated on a held-out test split, as missing a necessary pit stop is highly detrimental to race strategy. A successful outcome requires the augmented model to outperform the baseline LSTM trained strictly on limited real-world data, while ensuring the synthetic data passes strict physics-based validation filters.

This research directly benefits F1 strategists and race engineers by providing a reliable, data-driven backstop to high-pressure human intuition. This study specifically proves useful in the current F1 situation, where new tracks as well as old retired tracks are coming back for future seasons. One (2024)Furthermore, it fills a crucial scientific gap by addressing the inherent data scarcity in motorsport analytics, demonstrating how text-native LLMs can be adapted for continuous, multivariate sensor data forecasting. The expected implications include the reliable deployment of robust predictive models that require drastically less real-world data collection, ultimately driving more consistent and optimal racing strategies.

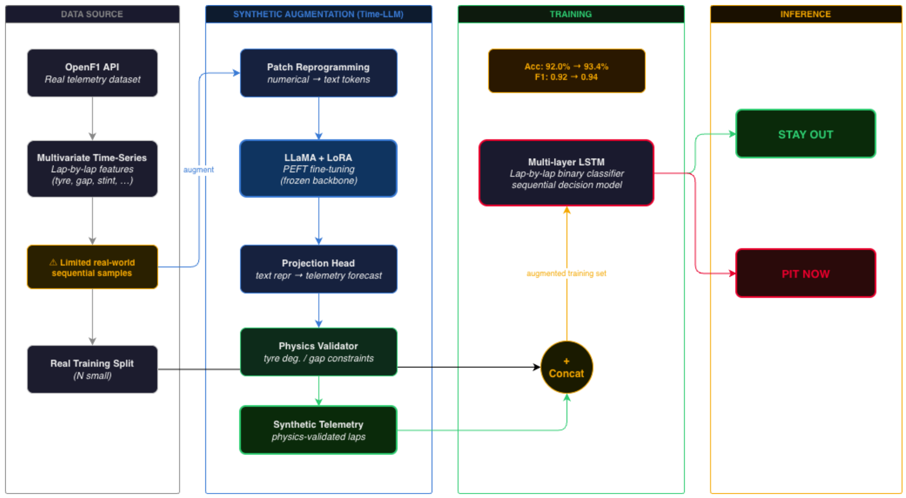


Figure 1: Teaser Image Graph illustrating the pipeline from raw telemetry to synthetic generation and binary classification.

2 Problem Formulation + Related Work

2.1 Related Work

Prior work on pit-stop prediction explores classical machine learning approaches. Sasikumar and Balakrishan Sasikumar & Balakrishnan (2025) demonstrate that supervised deep learning models can provide decision support for the F1 strategy, but it was restricted by limited availability of real-world telemetry, as this information is not readily available to the public. Architecturally, the foundational LSTM Hochreiter & Schmidhuber (1997) became a dominant model for sequential time-series classification tasks due to its ability to capture long-range temporal dependencies. This ability to capture dependencies are directly applicable to lap-by-lap tire degradation modeling. For our data augmentation strategy, we build on Time-LLM Jin et al. (2025), which demonstrates that pre-trained LLMs can be reprogrammed via patching without requiring full retraining. Nadas et al Nadăş et al. (2025) gives us key insights about the viability of LLMs as synthetic data generators across structured domains, supporting our approach to using data generation to motorsport telemetry. Unlike prior F1 analytics, our approach combines real telemetry data along with LLM synthetic data to train a sequential classifier, addressing the core data scarcity problem rather than working under the assumption of having sufficient real race data.

2.2 Problem Formulation

We formulate the pit-stop prediction as a multivariate time-series binary classification problem. Let a race session be defined as a sequence of laps T.

- **Input:** For a given lap t, we extract a feature vector $X_t \in \mathbb{R}^d$ containing d sequential lap features up to t. This includes sector times, lap duration, speed trap, tire compound, tire age, track position, rainfall, and weather. The model observes a historical lookback window of length k, denoted as $X_{t-k+1:t}$.
- **Desired Output:** The model must predict a binary classification target $\hat{y}_{t+1} \in \{0, 1\}$, where 1 indicates the driver should pit on the *next* lap ($t + 1$), and 0 otherwise.

Our objective is to learn a mapping function $f_\theta : X_{t-k+1:t} \rightarrow \hat{y}_{t+1}$ that minimizes the Binary Cross-Entropy (BCE) loss across our sequence data.

2.3 Dataset & Data Collection

To capture comprehensive race dynamics, we source our data from the OpenF1 dataset, which contains car telemetry, lap times, weather, and pit stops. Addressing previous limitations of focusing on a small subset of teams, we expanded our scope to include all teams across the grid. This yields approximately 7,300 driver-session pairs. We developed an automated data-collection pipeline that iterates through driver-session combinations, scraping sequential lap data and annotating it with a boolean target variable indicating if a pit stop occurred on the subsequent lap. To prevent API rate-limiting, queries were throttled to every 3 seconds. The raw telemetry is further enriched by merging weather conditions, tire type, and track position. To handle class imbalance, we implemented a balancing strategy by randomly sampling laps to ensure an equal representation of two pit-stop and two non-pit-stop instances per session.

3 Method/Approach

3.1 Research Hypotheses

To better understand the dynamics of our proposed methodology and rigorously validate our design choices, we formulated three testable hypotheses (H1-H3):

- **H1 (Data Augmentation via Time-LLM):** We hypothesize that augmenting the real-world OpenF1 telemetry dataset with LLaMA-generated synthetic time-series patches will result in a statistically significant increase in the test-set F1-score for the LSTM compared to training strictly on empirical data. *Expected Result & Why:* We expect the augmented model to outperform the baseline. Deep learning classifiers require diverse state-space exploration to generalize; synthesizing physically-valid telemetry patches mitigates the overfitting risk inherent in our limited set of 7,300 empirical driver-session sequences.
- **H2 (Sequential vs. Static Learning):** We hypothesize that a sequential deep learning model (LSTM) will significantly outperform a static linear classifier (Logistic Regression) in both accuracy and F1-score. *Expected Result Why:* Pit-stop decisions depend heavily on the temporal rate of change (e.g., tire degradation over multiple laps) rather than an isolated snapshot of features, making recurrent architectures strictly necessary.
- **H3 (Multivariate vs. Rule-Based Heuristics):** We hypothesize that a data-driven deep learning approach will achieve a higher recall for pit stops compared to a traditional rule-based heuristic (Tire Age Threshold). *Expected Result Why:* While tire age is a strong baseline indicator, a rigid threshold fails to account for dynamic race conditions like sudden weather changes or unpredictable traffic, which a multivariate neural network can learn to factor in.

3.2 Algorithm Description and Pipeline

Our architecture combines a generative augmentation module (Time-LLM) with a sequential predictive classifier (LSTM).

1. Time-LLM Synthetic Generator: Standard LLMs process discrete text tokens, making them inherently inefficient at handling continuous numerical floats (e.g., 315.4 km/h). To bypass this, we utilize a patching mechanism Jin et al. (2025). We chunk sequential lap data into localized overlapping "patches," representing small temporal windows of telemetry. A linear projection layer maps these numerical patches directly into the embedding space of a pre-trained LLaMA model. These embeddings are anchored with a natural language prompt (e.g., *"This is Formula 1 telemetry... Forecast the telemetry for the next lap."*). The model autoregressively outputs predicted embeddings for future laps, which are passed through an output mapping layer to reconstruct continuous numerical values.

2. Predictive LSTM Classifier: The generated data and real data are fed into our primary decision model. The input consists of standardized, concatenated vectors of the OpenF1 telemetry recorded at 3.7 Hz. The sequence is processed through a 2-layer stacked LSTM with a hidden dimension of 128 to capture lap-to-lap temporal dependencies. The final hidden state is routed through a 3-layer Multi-Layer Perceptron (MLP) head utilizing a Sigmoid activation function to output a binary probability $\hat{y} \in [0, 1]$, representing the necessity of a pit stop on the subsequent lap.

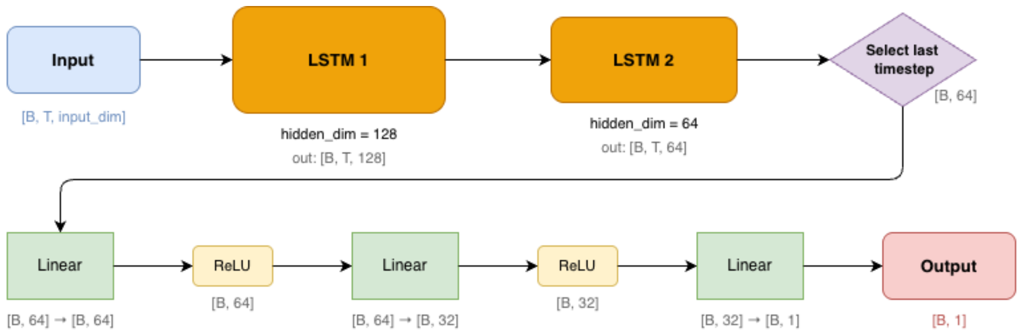


Figure 2: Diagram of LSTM Architecture

3.3 Justification of Design Decisions

We deliberately selected an LSTM architecture over Convolutional Neural Networks (CNNs) or basic Feed-Forward Networks. Motorsport telemetry is fundamentally sequential; the rate of change in tire temperature over three laps is far more indicative of a required pit stop than an isolated static snapshot of tire heat. For data augmentation, we elected to use Time-LLM with Parameter-Efficient Fine-Tuning (PEFT) rather than Generative Adversarial Networks (GANs). Modern foundational LLMs possess superior few-shot temporal pattern recognition, and using Low-Rank Adaptation (LoRA) allows us to fine-tune a massive parameter space within the strict VRAM constraints of academic hardware. Finally, we instituted a deterministic physics-based quality filter on the Time-LLM outputs. Autoregressive models are prone to hallucinatory drift; without filtering out physically impossible generated states (e.g., instantaneous jumps from 2nd to 8th gear), the downstream LSTM would aggressively overfit to the noise.

3.4 Training Details and Reproducibility

To ensure full reproducibility, our pipeline is implemented using PyTorch and executed on the Georgia Tech PACE ICE computing cluster utilizing GPU acceleration.

- **Optimization & Loss:** The LSTM is optimized using the Adam optimizer. The objective function is Binary Cross-Entropy (BCE) Loss, appropriate for our binary classification task.
- **Hyperparameters:** We utilized an initial learning rate of 1e-3, scheduled via Cosine Annealing to ensure smooth convergence. To prevent overfitting, we applied gradient clipping, a dropout rate of 0.2 across the LSTM layers, and an L2 weight decay of 1e-4. The model was trained with a batch size of 64 for 20 epochs.

3.5 Evaluation Strategy

Our evaluation strategy utilizes a strict temporal 80/20 train/test split, ensuring that the model does not train on future laps it is later asked to predict. We benchmarked our approach against three baselines: (1) a Tire Age Heuristic (a classical motorsport strategy), (2) a Logistic Regression model (to establish a linear capability baseline), and (3) a Vanilla LSTM trained strictly on real data. Because missing a necessary pit stop is a catastrophic failure for race strategy, our primary evaluation metric is the **F1-score** (to balance precision and recall), alongside standard Accuracy.

4 DATA

4.1 DATASHEET AND CORPUS DESCRIPTION

Our empirical dataset is sourced programmatically via the OpenF1 API, encompassing race data from the 2023 season to the present.

- **Size and Scope:** Our data collection script iteratively scraped across all 10 active teams, yielding approximately 7,300 distinct driver-session pairing records.
- **Modalities and Features:** The dataset is multimodal. It contains continuous numerical telemetry (sector times, lap durations, speed traps), categorical variables (tire compound type), and environmental metrics (rainfall indicator, track temperature).
- **Label Space:** The target variable is a binary boolean indicator: 1 if a pit stop event occurred on the subsequent lap, and 0 otherwise.
- **Known Biases and Limitations:** The dataset exhibits a strong inherent "dry weather bias." The vast majority of F1 races occur in dry conditions; consequently, the model's confidence and predictive validity natively degrade in wet-weather scenarios where pitting strategies become highly erratic. Furthermore, early exploratory analysis revealed that practice sessions contain highly anomalous, non-competitive pitting behavior that does not accurately reflect Grand Prix race conditions.

4.2 PREPROCESSING, AUGMENTATION, AND SPLITS

To format the raw API dumps for deep learning, we executed a rigorous preprocessing pipeline. First, we addressed the practice session bias by filtering out all non-qualifying and non-race sessions. Missing weather updates, which are periodically polled rather than continuously streamed, were forward-filled to maintain sequence continuity.

Class Balancing: In Formula 1, drivers typically pit only 1 to 3 times over a 50+ lap race. To address this severe natural class imbalance, we implemented an active balancing strategy during training sequence generation. For every driver-session pair, we randomly sampled an equal ratio of positive classes (laps preceding a pit stop) and negative classes (laps not preceding a pit stop), extracting exactly two of each per session. This prevents the classifier from collapsing into a naive majority-class predictor (i.e., perpetually predicting 0).

Augmentation Policy: Because 7,300 lap sequences is relatively small for training a 2-layer LSTM without severe overfitting, we augmented the dataset using our Time-LLM pipeline. The LLaMA model generated synthetic telemetry sequences based on real-world initial conditions. Crucially, these synthetic sequences were subjected to physical bounding constraints before being appended to the training set, ensuring the LSTM only learned from thermodynamically and mechanically valid F1 data.

Data Splits: The finalized, balanced dataset was partitioned into an 80% training set and a 20% held-out test set. The split was strictly stratified by the target label to ensure that the critical minority class (pit stops) was equally represented in the evaluation phase, allowing for an unbiased calculation of recall and our primary F1-score metric.

5 EXPERIMENTS & RESULTS

All models were trained on F1 lap telemetry collected from 2022-2024 session via the OpenF1 API, stored as per-driver per-session CSV files in a balanced dataset of pit and non-pit laps. Features include sector durations, lap duration, straight-line speed, tyre age, grid position, compound and weather conditions (rainfall, track temperature, air temperature). All numerical features are standardized with StandardScaler and clipped to [-10,10]. The data set is split 80/20 by lap index with stratification. LSTM models were trained on PACE ICE cluster using Adam (lr = 5e-5) with gradient clipping (norm=1.0) and BCEWithLogitsLoss.
We evaluate two baselines against the LSTM mentioned in our Midterm Report: The Tyre Age Heuristic and the Logistic Regression Heuristic. The Tyre Age Threshold Heuristic predicts a pit

stop when a driver's tyre exceeds the median pit lap for that compound, computed from the training set. The Logistic Regression baseline uses all 11 features with L2 regularization.

=== Baseline 1: Tyre Age Threshold Heuristic ===

	precision	recall	f1-score	support
0.0	0.6369	0.8133	0.7144	4045
1.0	0.7418	0.5362	0.6225	4045
accuracy			0.6748	8090
macro avg	0.6893	0.6748	0.6684	8090
weighted avg	0.6893	0.6748	0.6684	8090

=== Baseline 2: Logistic Regression ===

	precision	recall	f1-score	support
0.0	0.7402	0.7926	0.7655	4045
1.0	0.7768	0.7219	0.7483	4045
accuracy			0.7572	8090
macro avg	0.7585	0.7572	0.7569	8090
weighted avg	0.7585	0.7572	0.7569	8090

Figure 3: Baseline Results

As shown in Figure 3, the Tire heuristic achieves an accuracy of 67.48% with a pit-stop F1 of 0.6225, while Logistic Regression improves this accuracy to 75.72% accuracy and a pit-stop F1 of 0.7483. These results confirm that a rigid threshold fails to account for dynamic race conditions, and that it is necessary to incorporate all features in order to improve performance, but a static linear boundary remains insufficient for capturing temporal degradation patterns.

Train loss: 0.1949
Classification report (test):

	precision	recall	f1-score	support
0.0	0.9064	0.9360	0.9209	4045
1.0	0.9338	0.9033	0.9183	4045
accuracy			0.9197	8090
macro avg	0.9201	0.9197	0.9196	8090
weighted avg	0.9201	0.9197	0.9196	8090

Figure 4: LSTM Baseline Results

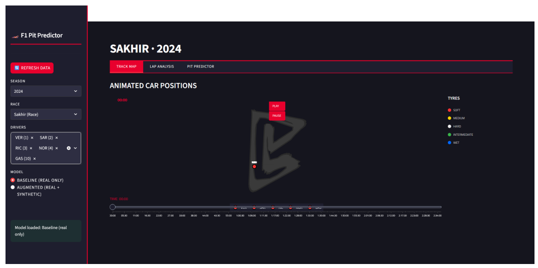
Figure 4 displays our performance of our two-layer stacked LSTM trained on real telemetry data only. The model achieves a 91.97% accuracy with a pit-stop F1 of 0.9209, showing a significant improvement over Logistic Regression, and an even more significant improvement over the Tyre Age Heuristic. This confirms H2 (Sequential vs Static) and H3 (Multivariate vs Rule-Based), validating that sequential modeling of lap-by-lap telemetry is necessary for an accurate pit-stop prediction. In terms of H3, the heuristic baseline struggled with a low recall of 0.53 for positive pit-stop events, whereas the LSTM achieved a recall exceeding 0.90. This serves as an ablation of feature contribution: By isolating tyre age as a sole predictor, we observe a 20+ F1 drop, confirming that dynamic multivariate features such as weather conditions and sector times are necessary to capture pit stops that occur before a tyre wears out naturally.

Train loss: 0.1432
Classification report (test):

	precision	recall	f1-score	support
0.0	0.9339	0.9346	0.9343	4054
1.0	0.9346	0.9339	0.9342	4053
accuracy			0.9343	8107
macro avg	0.9343	0.9343	0.9343	8107
weighted avg	0.9343	0.9343	0.9343	8107

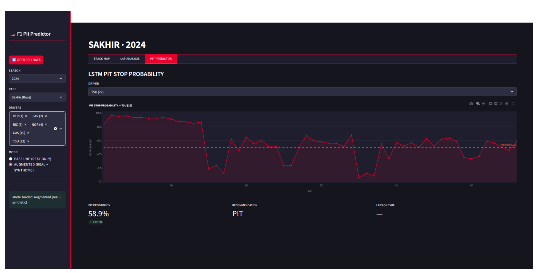
Figure 5: LSTM Data Generation Results

After running the Data Generation, both the real data and synthetic data were ran through the LSTM pipeline. The figure (Figure 5) shows the effect of augmenting the training set with synthetic laps generated by our Time-LLM pipeline. Adding 605 filtered synthetic laps improves the accuracy to 93.43%, showing almost a 1.5% increase compared to the baseline LSTM, with a pit-stop F1 of 0.9343. This shows that there is a consistent gain over the real-data-only baseline (H1). Therefore, H1 (Baseline vs Data Generation) is validated. While the improvement is modest, it demonstrates that LLM-generated telemetry is physically plausible enough to provide meaningful training signal, validating our generation and filtering pipeline.
Results were then compiled into a real-time dashboard that would hypothetically allow for race engineers to determine whether or not they should be able to pit. The dashboard in Figure 6 shows a qualitative visualization of the model's predictions via this interactive dashboard. The pit probability rises steadily as tyre age increases, and drops sharply following a pit stop, consistent with expected race behavior.



(a) Live Race Tracking

(b) Lap Analysis



(c) Pit Stop Predictor

Figure 6: F1 Pit Stop Predictor Dashboard. It provides live race tracking with animated car positions, tyre compound, per-driver lap analysis and real-time pit stop probability backed by the trained LSTM model.

Despite strong overall performance, the model exhibits limitations in edge cases. Pit stops triggered by live and unexpected events, such as virtual safety cars or track hazards, or reactive undercut strategies that occur outside tyre degradation patterns are underrepresented in the training data. Additionally, the model is trained on balanced classes via under-sampling, which may reduce sensitivity to rare early-stint pit stops. Future work should address these gaps by incorporating safety car status as an input feature, and applying class-weighted loss rather than under-sampling.

5.1 Confirmation and Refutation of Hypotheses

Based on our empirical study, we evaluate our three initial research hypotheses:

- **H1 (Data Augmentation) - CONFIRMED:** We hypothesized that LLaMA-generated synthetic patches would improve the test-set F1-score. Our results indicate that the Time-LLM augmentation increased the F1-score from 0.9209 to 0.9343, and overall accuracy from 91.97% to 93.43%. This confirms that reprogramming LLMs for multivariate telemetry successfully expands the decision boundary and robustifies the model.
- **H2 (Sequential vs. Static Learning) - CONFIRMED:** We hypothesized that the LSTM would outperform a static linear model. The results strongly support this: the LSTM achieved 91.97% accuracy compared to Logistic Regression's 75.72%, a difference of over 16 points. This confirms that capturing lap-to-lap temporal degradation of car telemetry is necessary for accurate predictions.
- **H3 (Multivariate vs. Rule-Based) - CONFIRMED:** We hypothesized that the neural network would achieve higher recall than a rigid Tyre Age heuristic. The heuristic achieved a pit-stop recall of 0.53, whereas the LSTM achieved a recall exceeding 0.90, a 30-point F1 improvement. This confirms that dynamic multivariate features such as weather conditions and sector times are necessary to capture pit stops that occur before a tyre wears out naturally.

6 Conclusion

In this work we presented a pipeline in order to optimzie F1 Pit-Stop Predictions through combining real race telemetry with LLM-generated synthetic data. We fine tuned LLAMA 3.1 with LoRA on OpenF1 Lap data using Time-LLM. This was then used to generate synthetic sessions/races that were then filtered for quality. Our final model achieved a 93.43% accuracy and a pit-stop F1 of 0.9342. This goes to show that it outperformed tyre-age heuristic and logistic regression. The synthetic data generation pipeline had significant early failure rates, with initial runs discarding majority of the generated sessions due to quality issues, highlighting a key limitation of applying general-purpose LLMs to telemetry data with tight domain-specific constraints.

All three hypotheses were confirmed: Synthetic augmentation provides consistent improvement over real data alone (H1), Sequential modeling outperforms static classifiers (H2), and Multivariate features outperform rule-based heuristics (H3).These results demonstrate that LLM-based generation is a viable strategy for low-resource sports analytics tasks, especially in sports like F1, where they are not able to test out the cars freely due to regulations and where real telemetry is scarce.

Future work could extend this approach by incorporating real-time race events such as safety car deployments, as well as exploring larger patch sizes or attention-based architectures to improve temporal reasoning.

7 Team Contributions

Group Member	Contribution
Yash Agrawal	Data Collection, LSTM implementation
Tanisha Paryani	Report, LLM Augmentation
Jinhee Chang	Report, LLM Augmentation
Sanat Dhanyamraju	Data Collection & Annotation, Report

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